



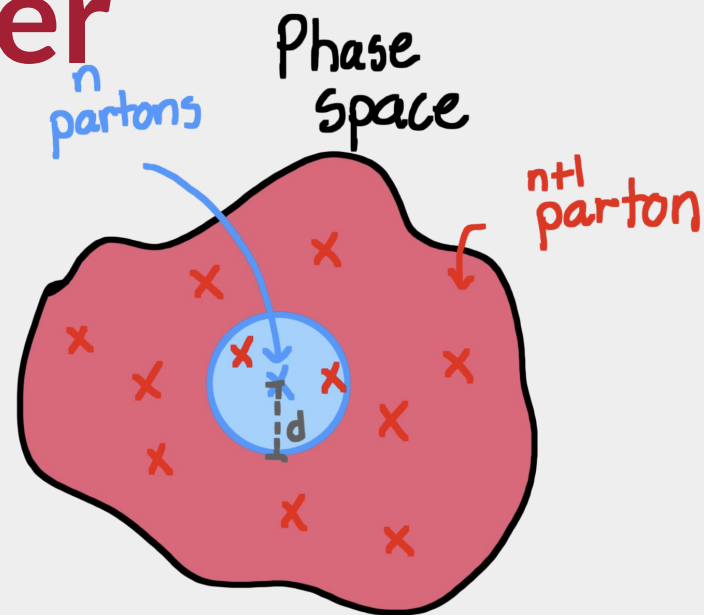
Positive Smeared Matrix Elements at Next-to-Leading Order

Rikab Gambhir

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In Collaboration with Andrew Larkoski

BOOST [Kraków] 2026





Positive Smeared Matrix Elements at

(next-to?)-

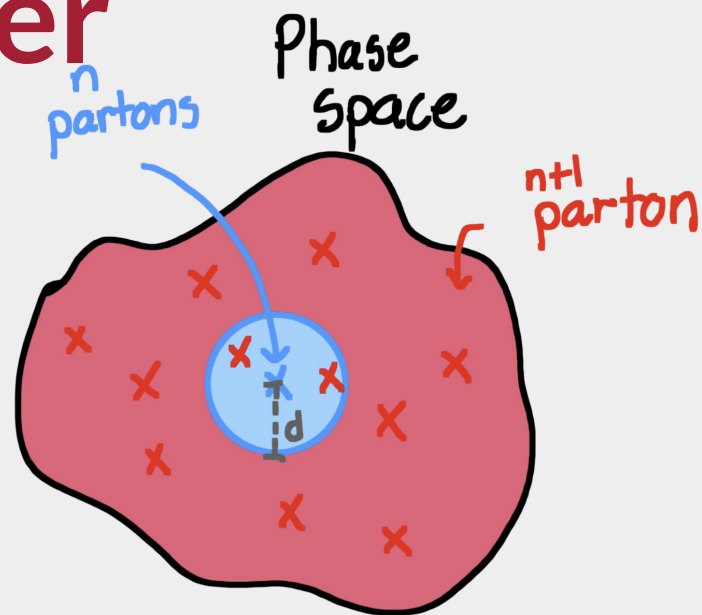
Next-to-Leading Order

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[Disclaimer] Everything I show today is highly **preliminary** and subject to change, especially NNLO. Comments welcome!

The Point

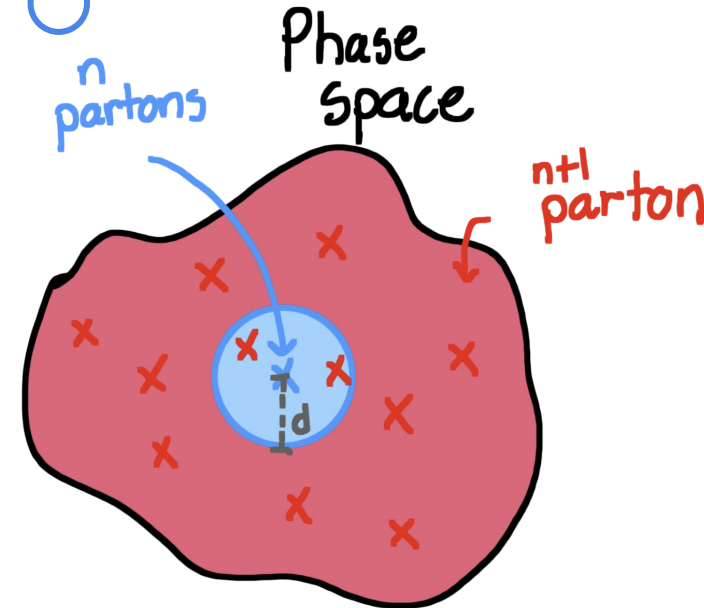
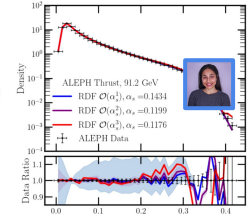
Fixed-order QCD at NLO produces cross sections that are **negative** and **divergent**.

This makes fixed-order event sampling difficult*, due to **large variance**.

Resummed Distribution Functions

Why should I care about this talk?

1. We introduce a new framework for **resummation** in QCD with a **single constraint of unitarity**
2. We provide a new **definition of theory uncertainties**
3. We perform a fit to ALEPH collider data to get a reasonable **measurement of α** using only **fixed-order calculations up to $O(\alpha^3)$**



$$\frac{\langle w^2 \rangle}{\langle w \rangle^2} = \frac{1}{(1 - 2f)^2}$$

Average effective number of events, if all weights are ± 1

We propose: **Smeared Matrix Elements**: In a sphere of radius d around negative events, absorb nearby real emissions into the matrix element *analytically*!

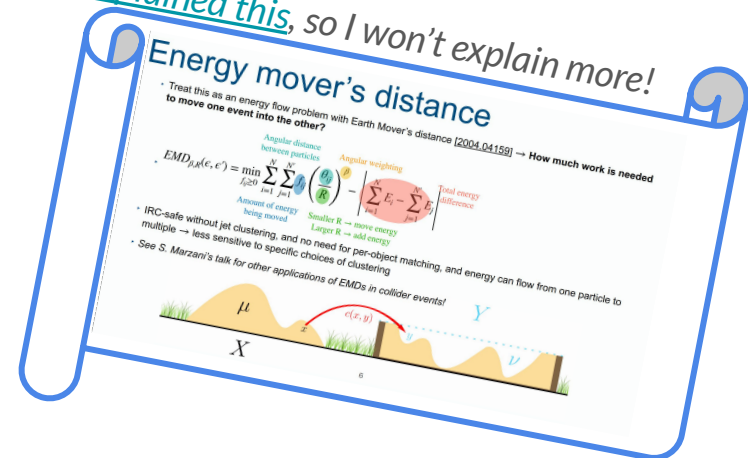
*[Footnote] Negative or divergent final cross sections are not an issue *within fixed-order perturbation theory*. These are real predictions of the theory. The issue is that these are *hard to sample*. That fixed-order perturbation theory is non-unitary is a separate issue, solved with by RDFs!

[Review] Metrics on QCD: (S)EMD

There are only a handful of genuinely perturbative metrics on QCD: the **(Spectral) Energy Mover's Distance**

The SEMD is useful because it can be computed in *closed form* for any N to M particle configuration, and in particular we can compute soft and collinear limits.

Jen just explained this, so I won't explain more!



Implemented using the **SPECTER** [package](#).



`pip install specterpy`

[Review] Metrics on QCD: (S)EMD

There are only a handful of genuinely

Introduced a few **BOOST**'s ago! (2024)

Jen just explained this, so I won't explain more!

Energy mover's distance

* Treat this as an energy flow problem with Earth Mover's distance [2004.04150] → How much work is needed to move one event into the other?

Angular distance between particles

$$d(\epsilon, \epsilon') = \min_{\phi} \sum_{i=1}^N \sum_{j=1}^{N'} \frac{\epsilon_i}{\epsilon_i} \frac{\epsilon'_j}{\epsilon'_j} \left(\frac{\theta_{ij}}{\epsilon_i} \right) - \sum_{i=1}^N \epsilon_i - \sum_{j=1}^{N'} \epsilon'_j$$

 Amount of energy being moved
 Smaller R → more energy
 Larger R → add energy
 Without jet clustering, and no need for per-object matching, and energy can flow from one object to another.
 It's talk for other applications of EMDs in color

The Spectral EMD: Closed Form

For $p = 2$, possible to find an *exact solution* for the optimal transport problem on the spectral representation of events:

$$\text{SEMD}_{\beta, p=2}(s_A, s_B) = \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min[S_A(\omega_n^+), S_B(\omega_l^+)] - \max[S_A(\omega_n^-), S_B(\omega_l^-)]) \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-))$$

Can be computed *exactly* in $O(N^2 \log N)^*$, as opposed to the full EMD in $O(N^3)$
 Closed form, easy derivatives and extremely easy to calculate programmatically!

Our framework for doing this, built in Python with JAX

SPECTER

See also: Sinkhorn, Sliced Wasserstein, WGANs, Linearized EMDs, ...

*A 1D OT is usually $O(K \log K)$, where K is the number of points. Here, $K \sim N^2$ for particle pairs

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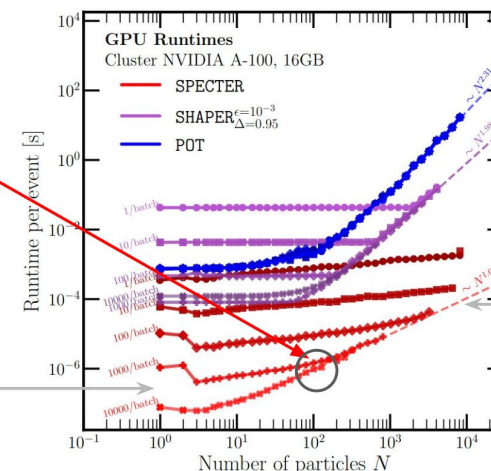


SPECTER is FAST (BOOSTED)!

Running on a single GPU on my local compute cluster ...

SEMDs between jets with ~100 particles can be computed at a rate of **one million SEMDs per second!**

Highly parallelized: Efficiency gains by processing many events at once!



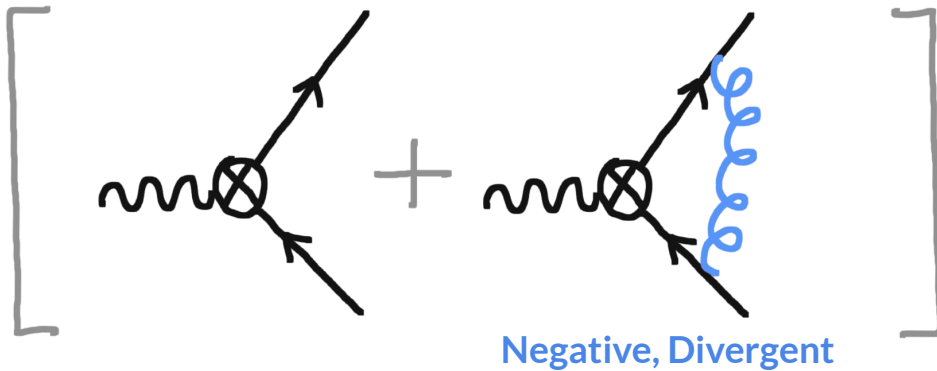
Worst case SEMD scaling is $O(N^2 \log N)$... SPECTER beats it in practical settings!

Rikab Gambhir – BOOST – 31 July 2024



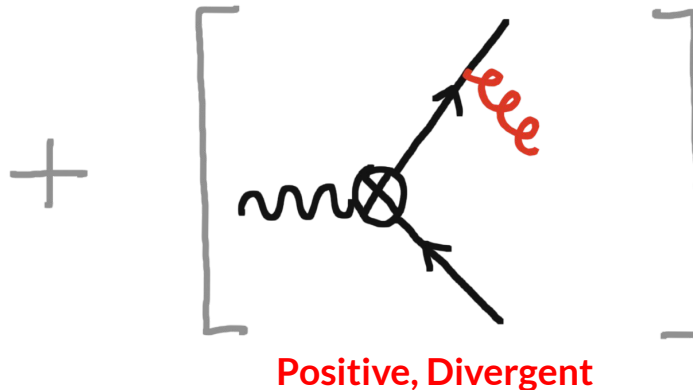
Anatomy of an NLO Calculation

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \Re \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) \right]$



Weights of n -parton events:
Negative and divergent

Resolved: Real Emission $\int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2$



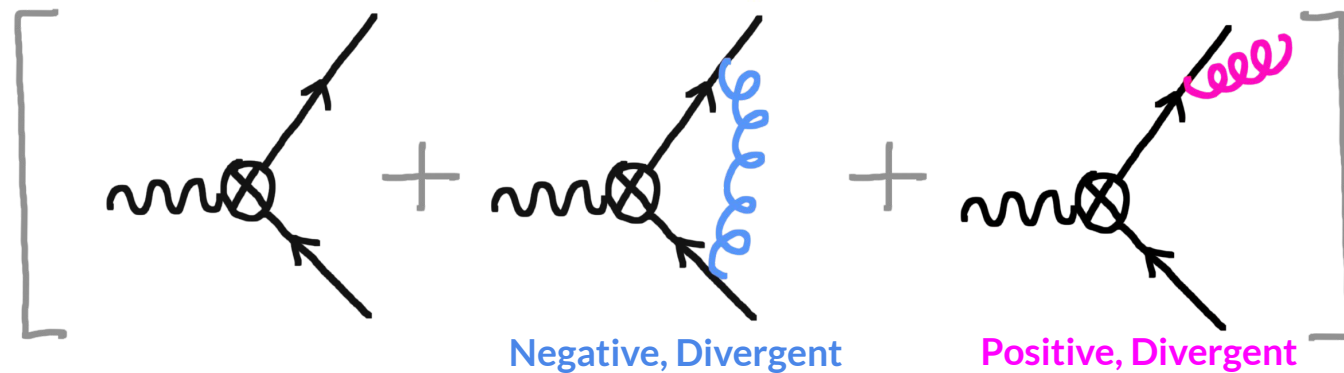
Weights of $n+1$ -parton events:
Positive and divergent

Anatomy of an NLO Calculation: Subtraction

Make each term finite by adding and subtracting the divergence. *Nothing has changed yet.*

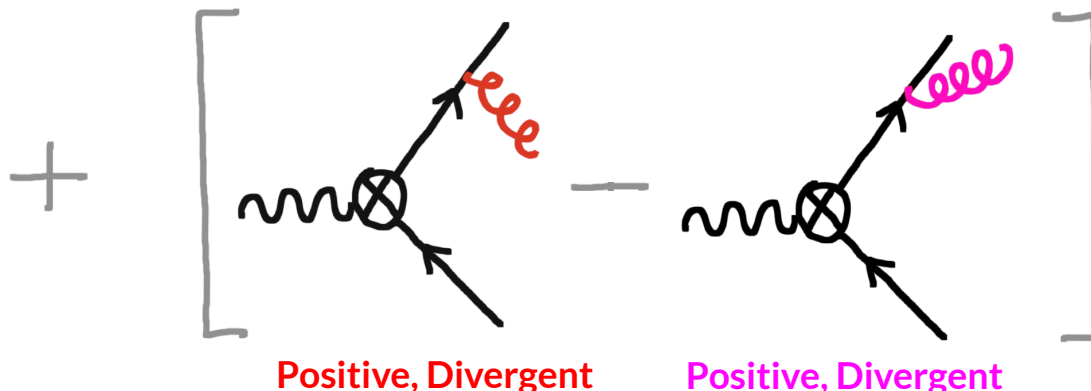
I'll choose to subtract phase space within a distance d of the Born event.

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \Re \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) \right] + \int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0^*))$



Pay attention to this, this is about to change.

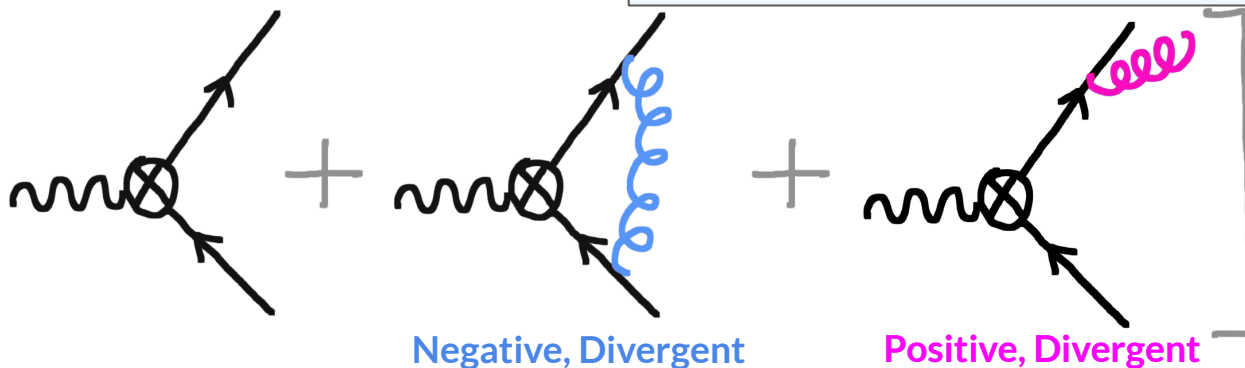
Resolved: Real Emission $\int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 - \int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0^*))$



Smearing

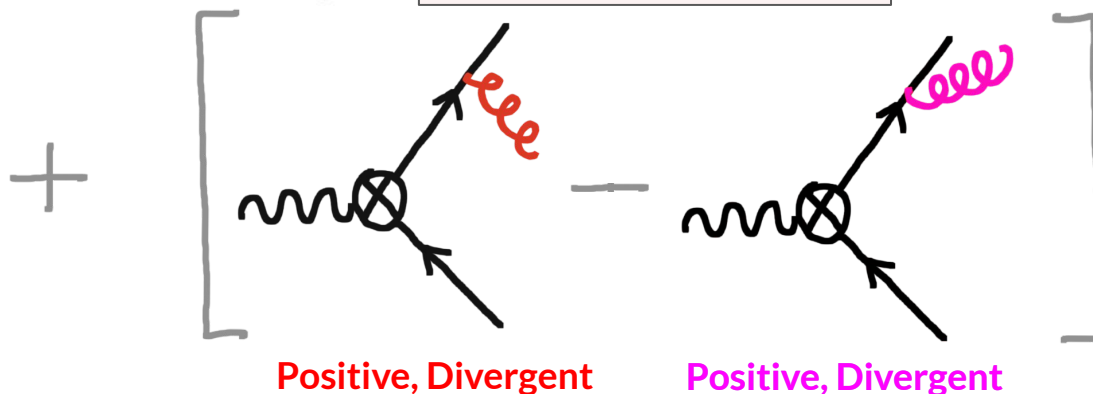
If a *resolved* event is close enough (within a distance d) of an unresolved event, treat it as contributing to the *total unresolved weight*.

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \operatorname{Re} \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + \int \frac{d\Pi_1}{d\Pi_0} \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0^*)) \right]$



Weights of n -parton events:
Finite and
interdeterminant sign

Resolved: Real Emission $\int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d(\Pi_1, \Pi_0^*) - d)$



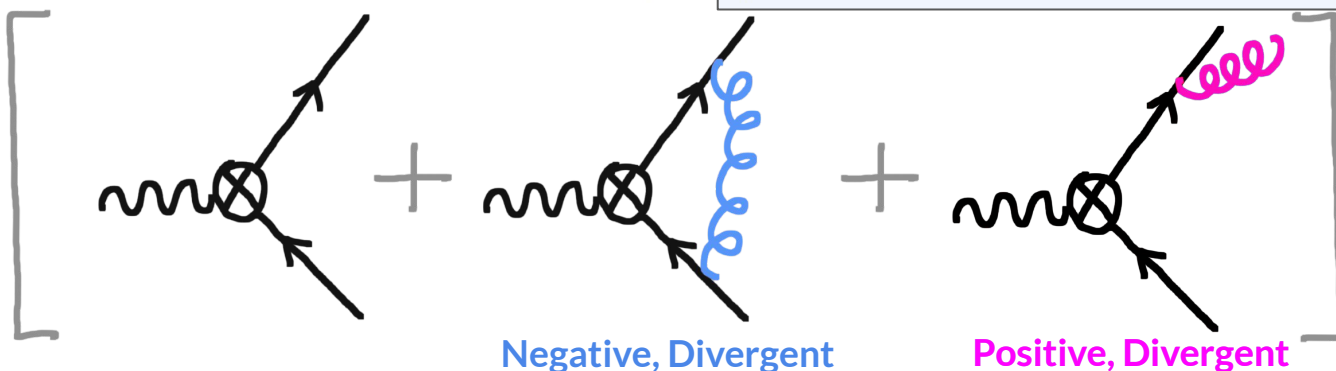
Weights of
 $n+1$ -parton events:
Finite and positive

Smearing

If a **resolved** event is close enough (within a distance d) of an unresolved event, treat it as contributing to the **total unresolved weight**.

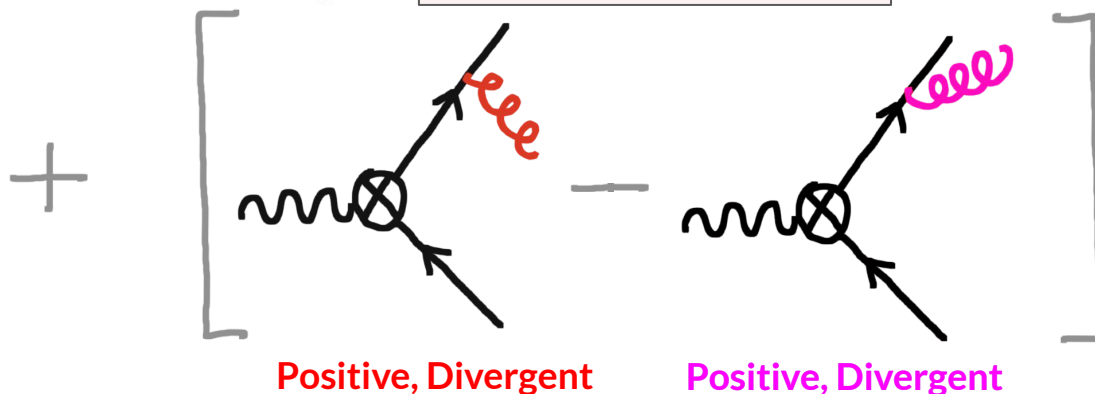
This *does* change physics because some $n+1$ parton events are now treated as n parton, but we claim the change is suppressed by $2(d/Q)^2$

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \operatorname{Re} \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + \int \frac{d\Pi_1}{d\Pi_0} \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0)) \right]$



Weights of n -parton events:
Finite and
interdeterminant sign

Resolved: Real Emission $\int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d(\Pi_1, \Pi_0^*) - d)$



Weights of
 $n+1$ -parton events:
Finite and positive

Smearing [Details]

If a *resolved* event is close enough (within a distance d) of an unresolved event, treat it as contributing to the *total unresolved weight*.

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \operatorname{Re} \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + \int \frac{d\Pi_1}{d\Pi_0} \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0)) \right]$

Step 1: Calculate this in QCD — doable in the soft/collinear limits for small d ! and for some processes, exactly!

Step 2: Generate n -parton phase space

Step 3: Apply this d -dependent weight

Will be negative for small d , positive for large d , and $\sim$ zero for d just right!

Weights of n -parton events:
Finite and
interdeterminant sign

Resolved: Real Emission $\int d\Pi_1 \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d(\Pi_1, \Pi_0^*) - d)$

Step 4: Generate $n+1$ parton phase space

Step 5: Cut away events below d — requires minimizing d between the given event and all n -parton events

Step 6: Done! The result is NLO events with weights correct up to $\sim (d/2Q)^2$. Standard unweighting can be applied.

Weights of $n+1$ -parton events:
Finite and positive

Smeared Matrix Elements

We can just do this once and for all (at least, the log part!)

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \operatorname{Re} \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + \int \frac{d\Pi_1}{d\Pi_0} \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0)) \right]$

Step 1: Calculate this in QCD – doable in the soft/collinear limits for small d ! and for some processes, exactly!

Weights of n -parton events:
Finite and
interdeterminant sign

For small d , the real emission piece factorizes into collinear and soft pieces:

$$\begin{aligned} & \lim_{d \rightarrow 0} \int \frac{d\Pi_1}{d\Pi_0} \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0)) \\ &= \left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 \int \frac{d\Pi_1}{d\Pi_0} \left[\sum_{i \in \Pi_0} P_i(z, s) + \sum_{i, j \in \Pi_0} \mathbf{T}_i \cdot \mathbf{T}_j \mathcal{A}_{ij}(p_s) \right] \Theta(d - d(\Pi_1, \Pi_0)) \end{aligned}$$

Smeared Matrix Elements

We can just do this once and for all (at least, the log part!)

Unresolved: Born + Virtual $\int d\Pi_0 \left[\left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \operatorname{Re} \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + \int \frac{d\Pi_1}{d\Pi_0} \left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 \Theta(d - d(\Pi_1, \Pi_0)) \right]$

Step 1: Calculate this in QCD – doable in the soft/collinear limits for small $d!$ and for some processes, exactly!

Weights of n -parton events:
Finite and
interdeterminant sign

Includes the virtual, soft, and collinear pieces.

$$\begin{aligned} \frac{|\mathcal{M}_0(\Pi_0|d)|^2}{|\mathcal{M}^{(0)}(\Pi_0)|^2} &= 1 - \frac{\alpha_s}{2\pi} (N_q C_F + N_g C_A) \log^2 \frac{Q^2}{2d^2} \\ &+ \frac{\alpha_s}{2\pi} \left[\frac{3}{2} N_q C_F + \left(\frac{11}{6} - \frac{2}{3} \frac{n_f T_R}{C_A} \right) N_g C_A + 2 \sum_{i \neq j} \mathbf{T}_i \cdot \mathbf{T}_j \log \frac{s_{ij}}{Q^2} \right] \log \frac{Q^2}{2d^2} \\ &- 4 \frac{\alpha_s}{2\pi} \sum_{i \in \Pi_0} C_i \log \frac{\sqrt{\frac{Q+2E_i}{4E_i}} + \mathcal{V}_i + \sqrt{\frac{Q+2E_i}{4E_i}} - \mathcal{V}_i}{2} \log \frac{Q^2}{2d^2} + \text{non-log terms.} \end{aligned}$$

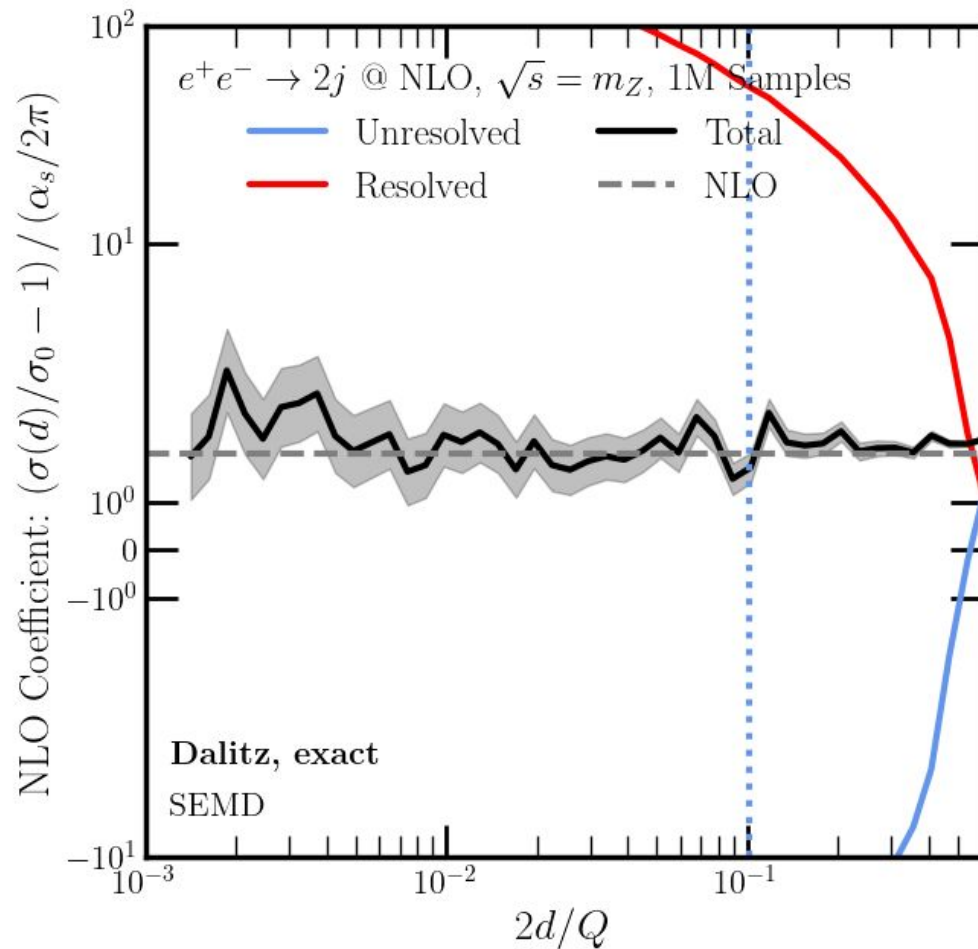
Born Level

$\mathcal{V}_k = \frac{1}{2E_k} \sqrt{\left(\frac{Q-2E_k}{2} \right)^2 - 2 \sum_{i,j \neq k \in \Pi_0} E_i E_j \cos^2 \frac{\theta_{ik}}{2} \cos^2 \frac{\theta_{jk}}{2} \sin^2(\phi_{ik} - \phi_{jk})}$

Kinematics, related to azimuthal angles

Non-universal terms

Proof of Concept: 2 Jets @ NLO

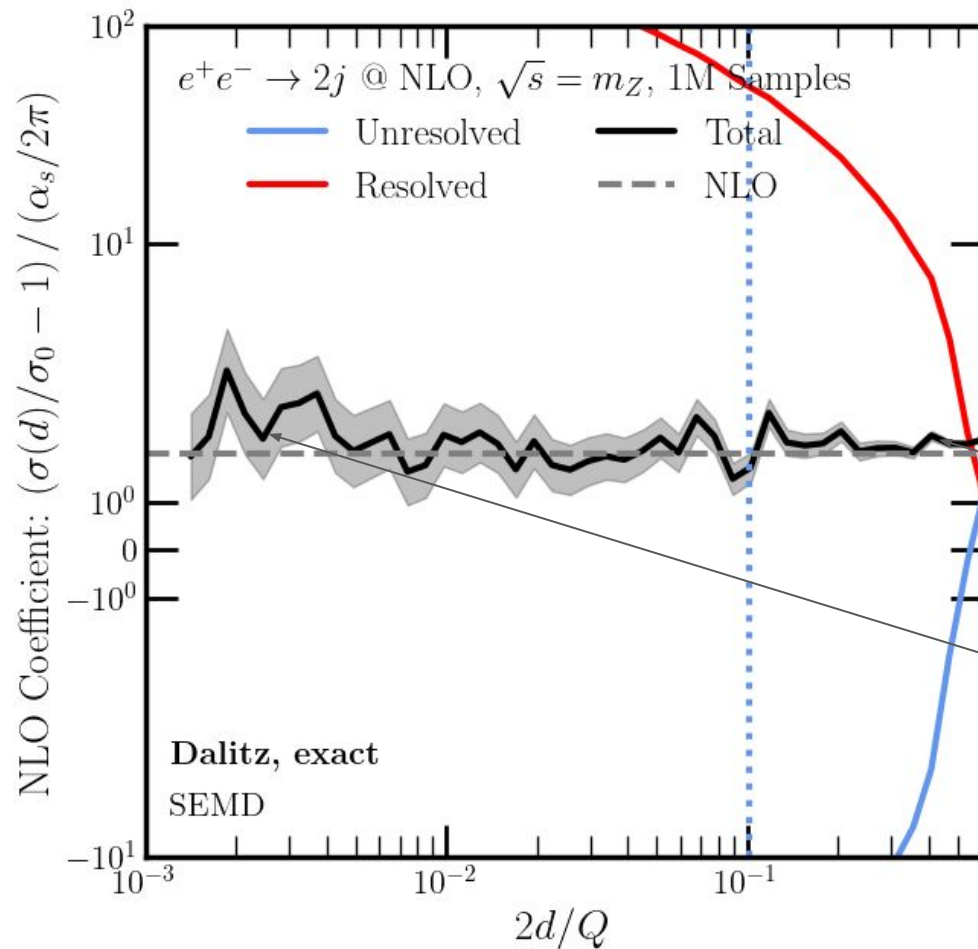


Total NLO 2 jet cross section as calculated with weighted events

Here, the “non-universal” piece is an easy-to-calculate numerical constant.

Here, only 1M samples are used per d to show exaggerated errors. Took ~seconds to do!

Proof of Concept: 2 Jets @ NLO



Total NLO 2 jet cross section as calculated with weighted events

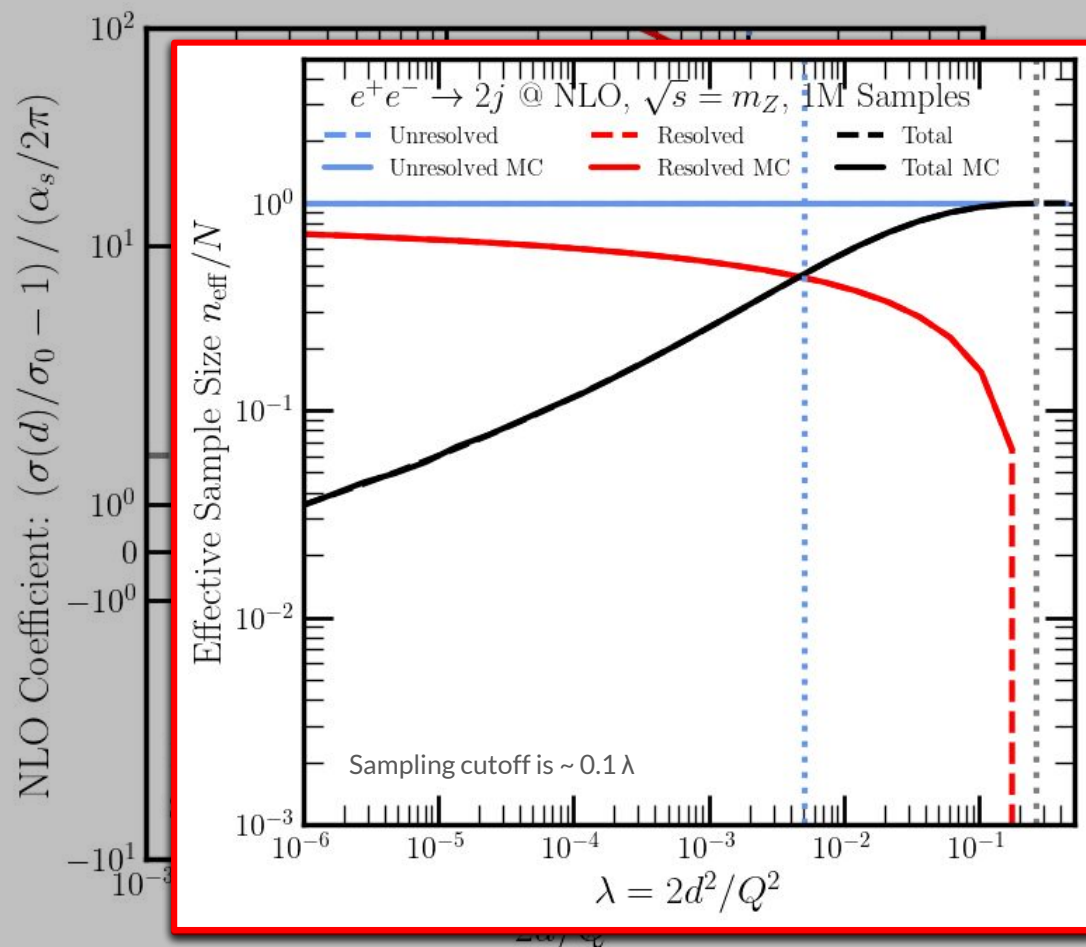
Here, the “non-universal” piece is an easy-to-calculate numerical constant.

Large d : all events are 2 parton events, slightly biased

Small d : Large logs, both positive and negative, poor cancellation

Proof of Concept: 2 Jets @ NLO

Statistical Power



We can compute an “Effective Sample Size” characterising the variance of the weights

Scales like

$$1 - \frac{11}{24L} + \mathcal{O}(1/L^2)$$

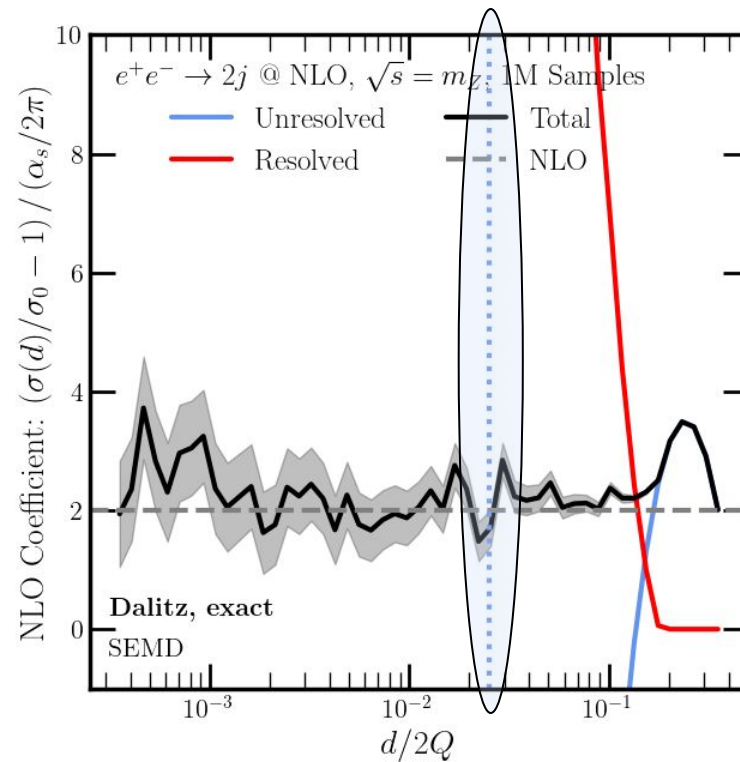
Approx 75% at the critical radius (defined shortly!)

ESS = Number of unweighted events with same variance

These technically depend strongly on the choice of phase space sampler. This uses the “efficient phase space sampler” from Stefan Hoeche’s notes: [1411.4085](https://arxiv.org/abs/1411.4085)

[Aside] The Critical Radius

We can estimate the minimum radius at which the virtual contribution is strictly positive



The blue curve doesn't cross 0 at the vertical line solely because the units on this plot are weird: NLO coefficient rather than total cross section

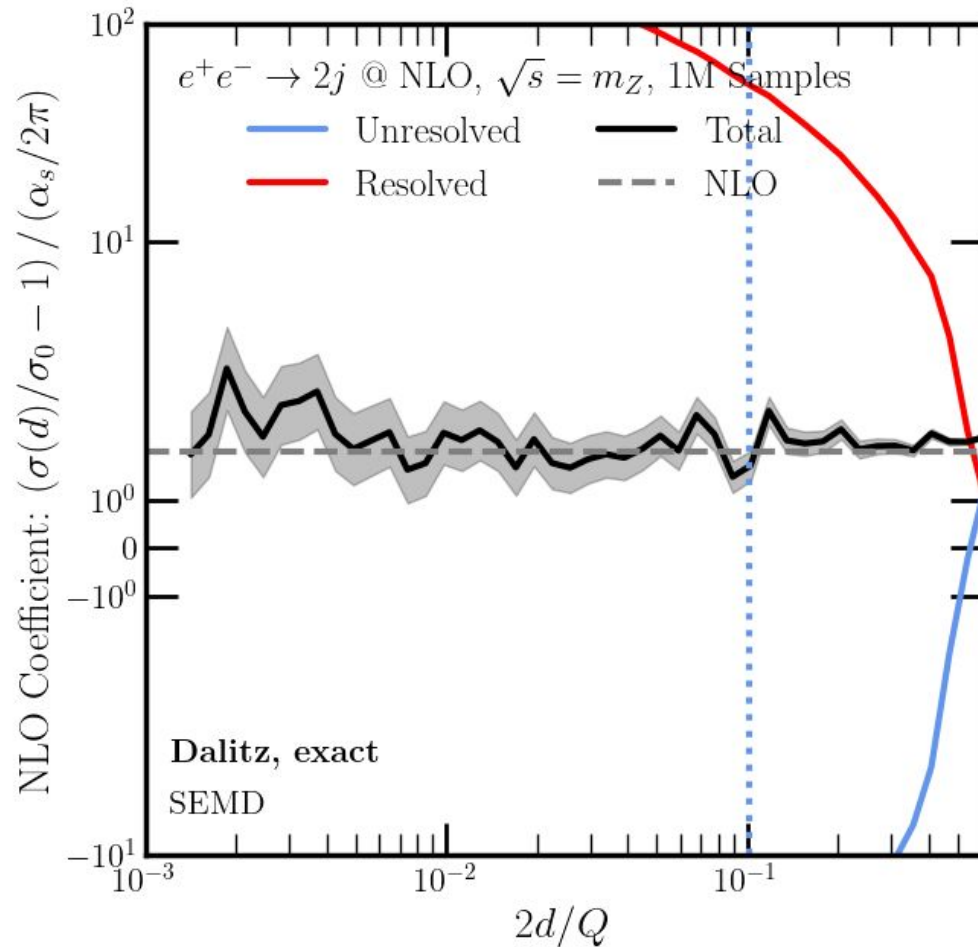
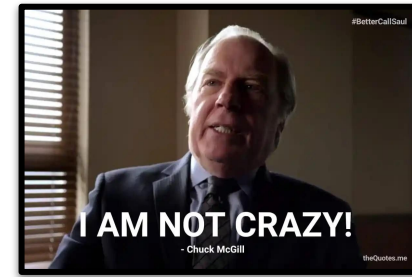
$$\log \frac{Q^2}{2d_{\text{crit}}^2} \equiv L_{\text{crit}} = \frac{1}{\sqrt{\alpha_s}} \sqrt{\frac{2\pi}{N_q C_F + N_g C_A}} + \frac{\frac{3}{4} N_q C_F + N_g \left(\frac{11}{12} C_A - \frac{n_f T_R}{3} \right) + \sum_{i \neq j} \mathbf{T}_i \cdot \mathbf{T}_j \log \frac{s_{ij}}{Q^2} - 2 \sum_{i \in \Pi_0} C_i \log \frac{\sqrt{\frac{Q+2E_i}{4E_i} + \mathcal{V}_i} + \sqrt{\frac{Q+2E_i}{4E_i} - \mathcal{V}_i}}{2}}{N_q C_F + N_g C_A} + \mathcal{O}(\sqrt{\alpha_s}).$$

For 2 Jets @ NLO at the Z pole:

$$d_{\text{crit}}^2 \approx \frac{m_Z^2}{600}$$

Positive and low variance weights with only a tiny regulator!

Proof of Concept: 2 Jets @ NLO

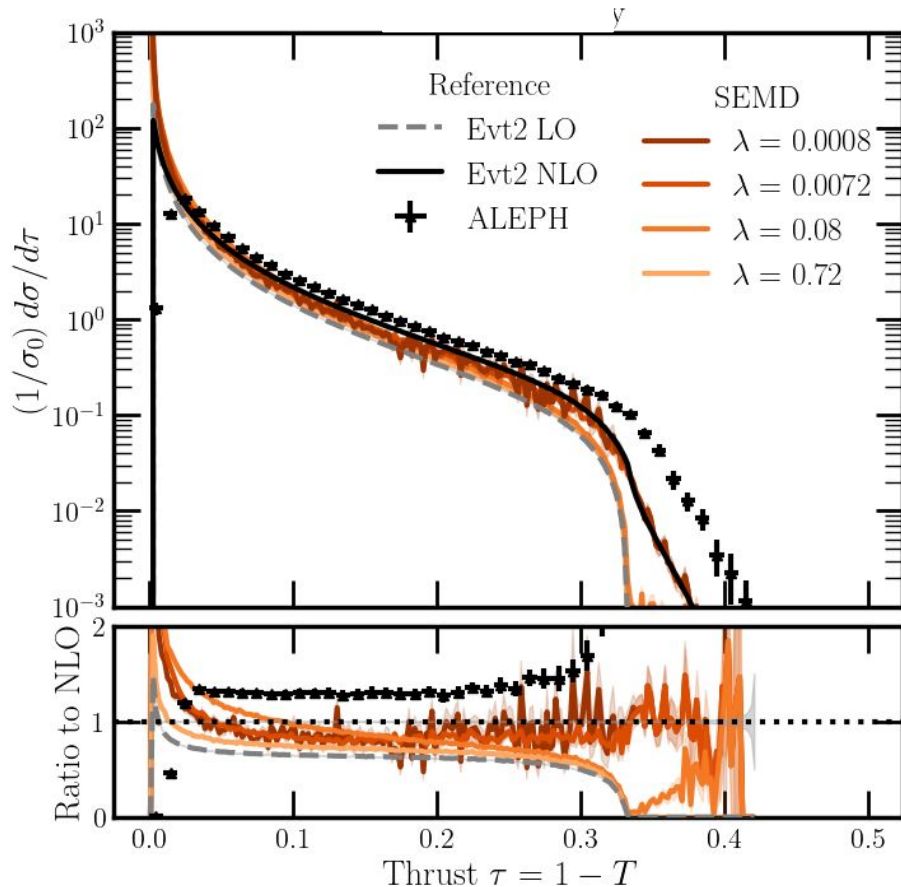


Total NLO 2 jet cross section as calculated with weighted events

Here, the “non-universal” piece is an easy-to-calculate numerical constant.

It works!

3 Jets @ NLO: Thrust

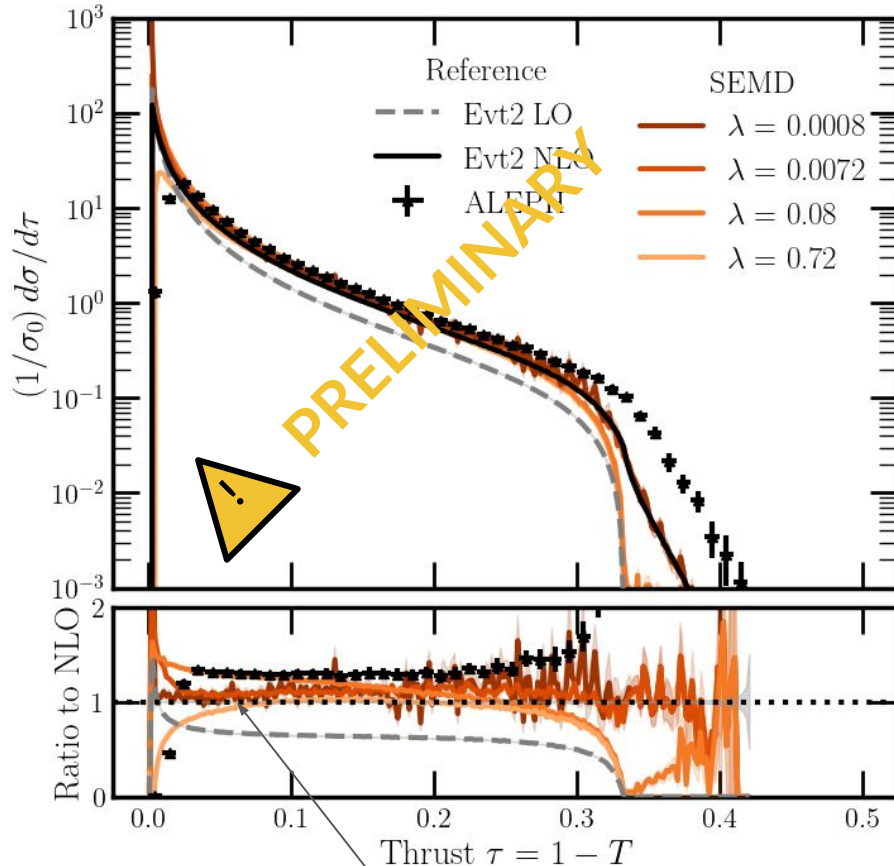


We can play a similar game with 3 jets at NLO. For comparison, we show EVENT2

! This is *only* the universal part of the smearing, so the match will not be perfect.

Small d to Large d interpolates nicely between NLO and LO

3 Jets @ NLO: Non-Universal Terms Calculable!



Compare to previous slide!

Virtual Piece

$$2\Re \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) = \left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 \frac{\alpha_s}{2\pi} \frac{1}{\Gamma(1-\epsilon)} \left(\frac{\mu^2}{Q^2} \right)^\epsilon \left[-\frac{1}{\epsilon^2} ((2C_F - C_A)y_{12}^{-\epsilon} + C_A(y_{13}^{-\epsilon} + y_{23}^{-\epsilon})) - \frac{1}{\epsilon} \left(3C_F + \frac{11}{6}C_A - \frac{2}{3}n_f T_R \right) + \frac{\pi^2}{2}(2C_F + C_A) - 8C_F \right] + \frac{16\pi^2}{Q^2} \left(\frac{\alpha_s}{2\pi} \right)^2 C_F F(y_{12}, y_{13}, y_{23})$$

Collinear Piece

$$\frac{\text{Coll}_{e^+e^- \rightarrow q\bar{q}g}}{|\mathcal{M}^{(0)}(\Pi_0)|^2} = \frac{\alpha_s C_F}{2\pi} \left(\frac{\mu^2}{2d^2} \right)^\epsilon \left(\frac{1+\sqrt{x_1}}{2\sqrt{x_1}} \right)^{2\epsilon} \left[\frac{2}{\epsilon^2} + \frac{3}{2}\frac{1}{\epsilon} + \frac{7}{2} - \frac{\pi^2}{2} + 4\text{Li}_2 \left(\frac{1-\sqrt{x_1}}{1+\sqrt{x_1}} \right) \right] + \frac{\alpha_s C_F}{2\pi} \left(\frac{\mu^2}{2d^2} \right)^\epsilon \left(\frac{1+\sqrt{x_2}}{2\sqrt{x_2}} \right)^{2\epsilon} \left[\frac{2}{\epsilon^2} + \frac{3}{2}\frac{1}{\epsilon} + \frac{7}{2} - \frac{\pi^2}{2} + 4\text{Li}_2 \left(\frac{1-\sqrt{x_2}}{1+\sqrt{x_2}} \right) \right] + \frac{\alpha_s C_A}{2\pi} \left(\frac{\mu^2}{2d^2} \right)^\epsilon \left(\frac{1+\sqrt{x_3}}{2\sqrt{x_3}} \right)^{2\epsilon} \left[\frac{2}{\epsilon^2} + \frac{1}{\epsilon} \left(\frac{11}{6} - \frac{2}{3} \frac{n_f T_R}{C_A} \right) + \frac{67}{18} - \frac{\pi^2}{2} - \frac{13}{9} \frac{n_f T_R}{C_A} + 4\text{Li}_2 \left(\frac{1-\sqrt{x_3}}{1+\sqrt{x_3}} \right) \right]$$

Soft Piece

$$\frac{\text{Soft}_{e^+e^- \rightarrow q\bar{q}g}}{|\mathcal{M}^{(0)}(\Pi_0)|^2} = \frac{\alpha_s}{2\pi} \left(\frac{\mu^2 Q^2}{4d^4} \right)^\epsilon \left[-\sum_{i \in \Pi_0} C_i \left(\frac{1+\sqrt{x_i}}{2} \right)^{4\epsilon} \left(\frac{1}{\epsilon^2} + \frac{\pi^2}{12} + 8\text{Li}_2 \left(\frac{1-\sqrt{x_i}}{1+\sqrt{x_i}} \right) + 4\text{Li}_2 \left(-\frac{1-\sqrt{x_i}}{1+\sqrt{x_i}} \right) \right) + \left(\frac{C_A}{2} - C_F \right) \left(\frac{x_1^{2\epsilon} + x_2^{2\epsilon}}{\epsilon} \log \frac{1-x_3}{x_1 x_2} - 2\text{Li}_2 \left(\frac{(1-x_1)(1-x_2)}{x_1 x_2} \right) + \log^2 \frac{1-x_3}{x_1 x_2} + \frac{I_\epsilon^{(12)}}{\pi} \right) - \frac{C_A}{2} \left(\frac{x_1^{2\epsilon} + x_3^{2\epsilon}}{\epsilon} \log \frac{1-x_2}{x_1 x_3} - 2\text{Li}_2 \left(\frac{(1-x_1)(1-x_3)}{x_1 x_3} \right) + \log^2 \frac{1-x_2}{x_1 x_3} + \frac{I_\epsilon^{(13)}}{\pi} \right) - \frac{C_A}{2} \left(\frac{x_2^{2\epsilon} + x_3^{2\epsilon}}{\epsilon} \log \frac{1-x_1}{x_2 x_3} - 2\text{Li}_2 \left(\frac{(1-x_2)(1-x_3)}{x_2 x_3} \right) + \log^2 \frac{1-x_1}{x_2 x_3} + \frac{I_\epsilon^{(23)}}{\pi} \right) \right]$$

Annoying finite pieces due to the SEMD in the soft limit. This part is still a WIP, hence the

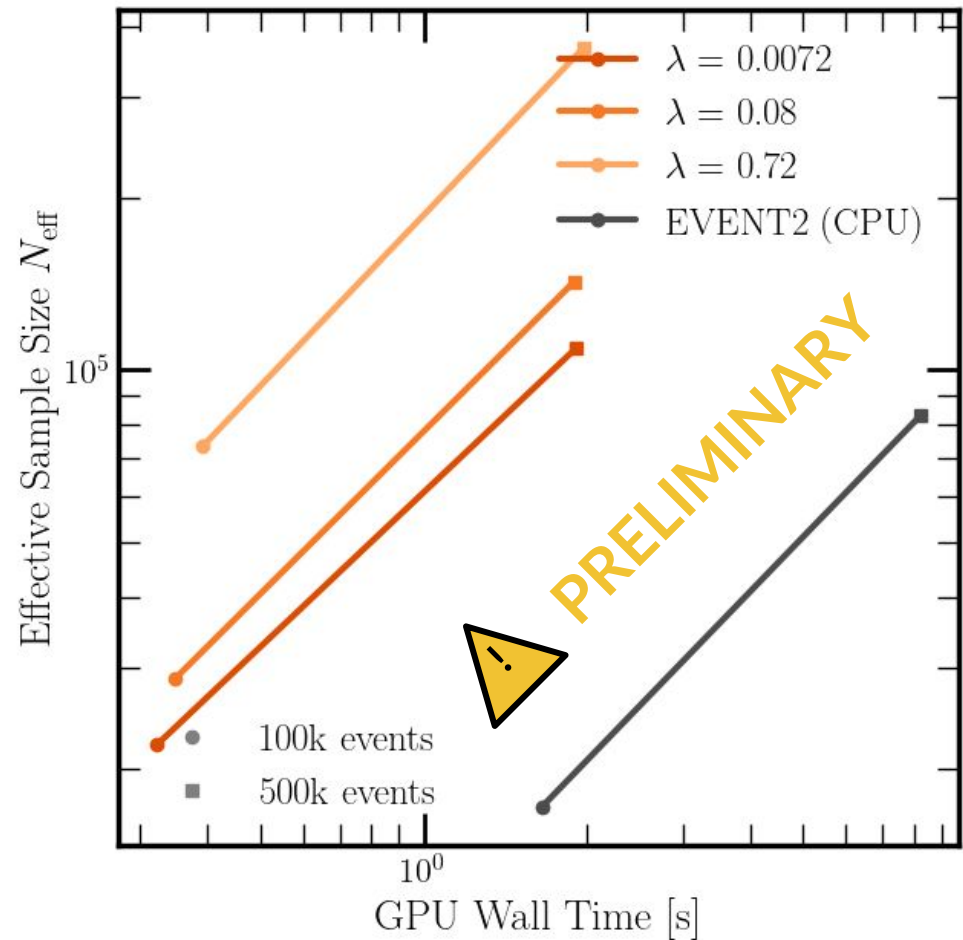
Time Comparison

Comparing the runtime per *effective sample size*, since that is the actual quantity of interest.

At low d , **approximately 5x more efficient** than EVENT2!

[Details] Calculated as the weight variance of the total cross section between a thrust of 0.05 and 0.35.

Runtime will likely improve in the final version — stay tuned :)



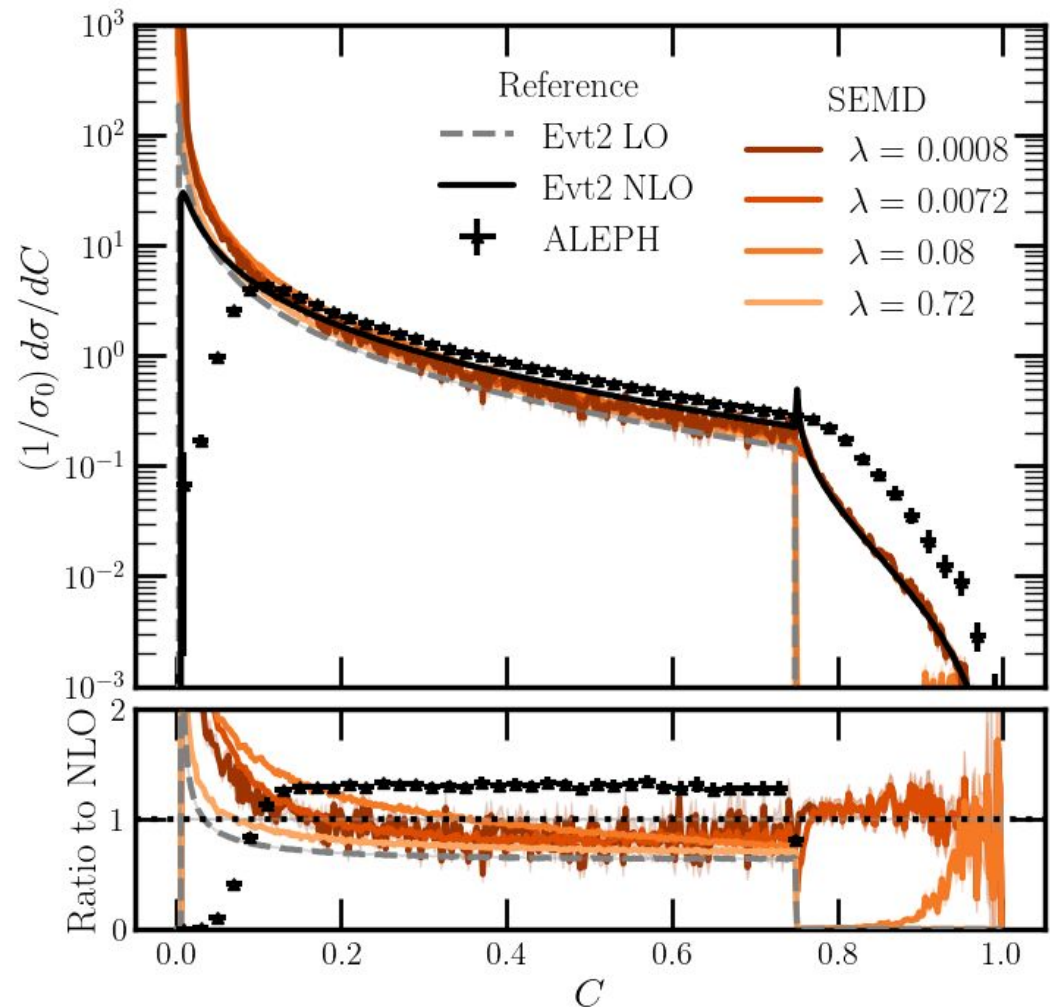
For those who care: Implemented in JAX, as all Python code should be!

3 Jets @ NLO: C-Parameter

We even reproduce the Sudakov shoulder at small d !

Finite d acts as a regulator.

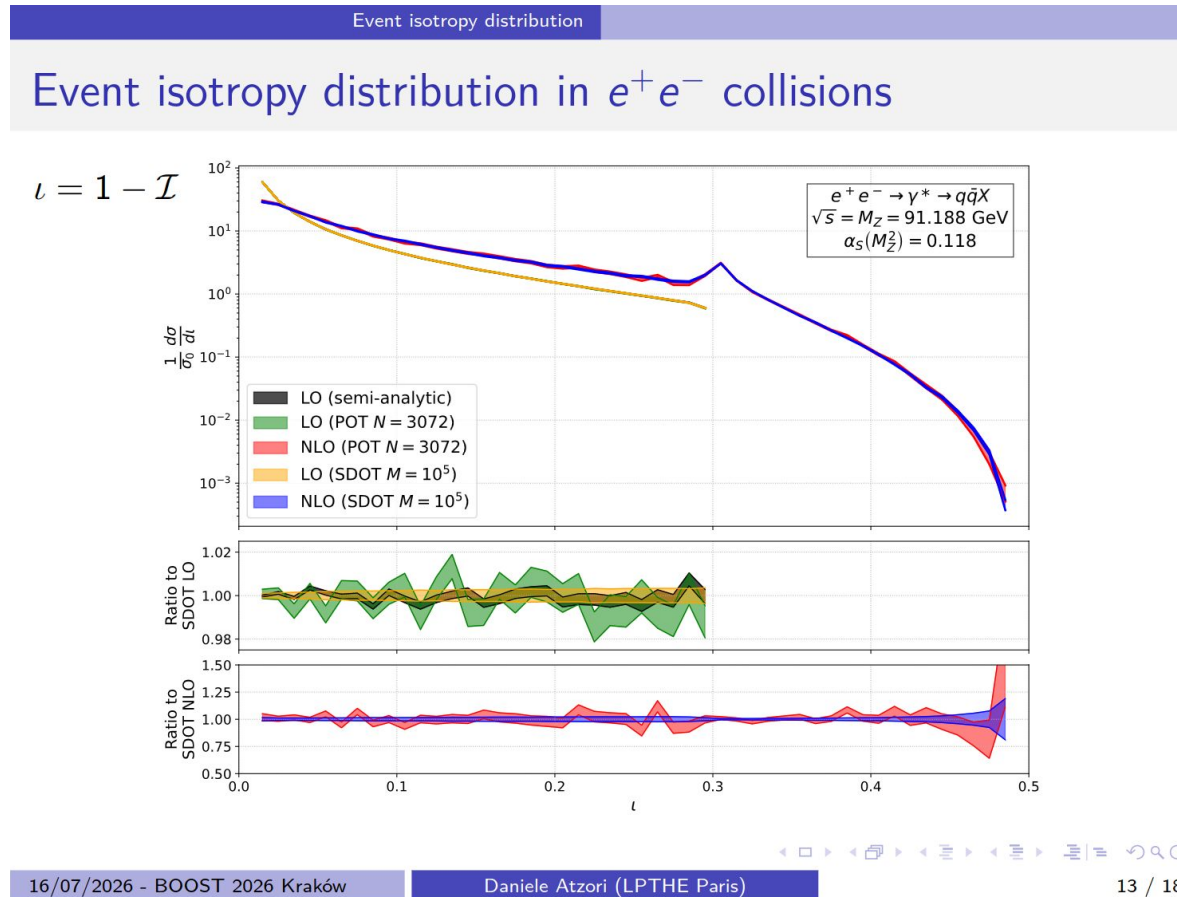
This plot: Universal logs only



3 Jets @ NLO: Just for fun, since we're at BOOST

Event Isotropy

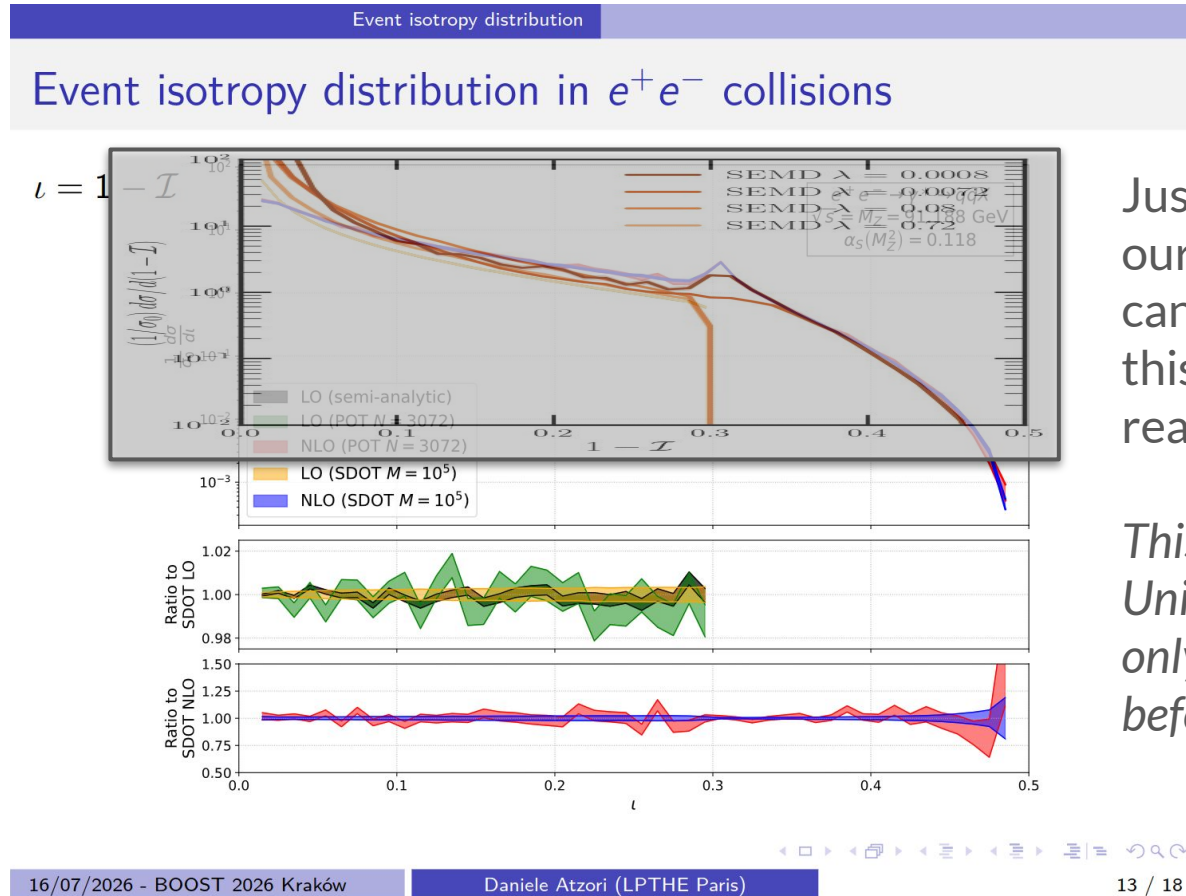
From [Daniele Atzori's Talk](#), earlier today!



3 Jets @ NLO: Just for fun, since we're at BOOST

Event Isotropy

From [Daniele Atzori's Talk](#), earlier today!



Just checking if our generator can reproduce this! Seems reasonable.

This plot: Universal logs only, made ~1hr before this talk.

2 Jets @ NNLO



At NNLO, we have to assign 2-parton, 3-parton, and 4-parton weights.

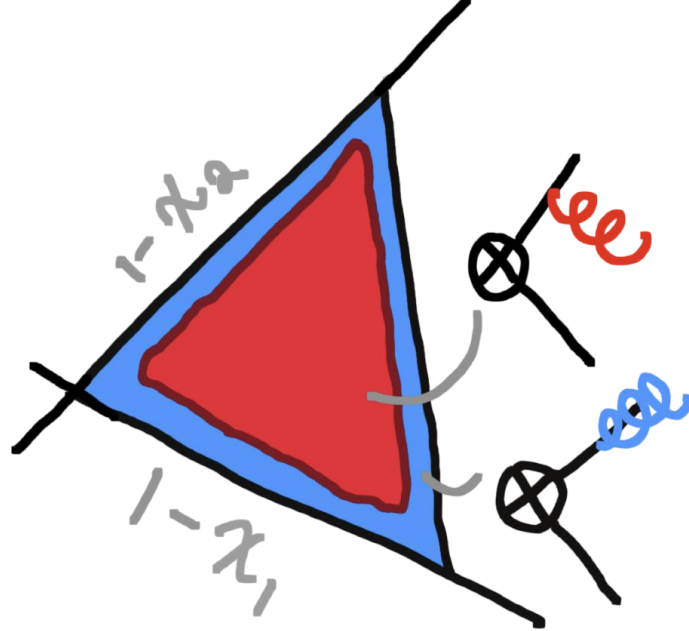
Two distance scales: 3-to-2 (d_0) and 4-to-3 (d_1).

$$\begin{aligned}
 |\mathcal{M}_0(\Pi_0|d_0)|^2 &= \left| \mathcal{M}^{(0)}(\Pi_0) \right|^2 + 2 \Re \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + 2 \Re \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_{VV}^{(2)}(\Pi_0)^* \right) \\
 &\quad + \int \frac{d\Pi_1}{d\Pi_0} \left[\left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 + 2 \Re \left(\mathcal{M}_R^{(1)}(\Pi_1) \mathcal{M}_{RV}^{(2)}(\Pi_1)^* \right) \right] \Theta(d_0 - d(\Pi_1, \Pi_0)) \\
 &\quad + \int \frac{d\Pi_2}{d\Pi_0} \left| \mathcal{M}_{RR}^{(2)}(\Pi_2) \right|^2 \Theta(d_0 - d(\Pi_2, \Pi_0)) , \\
 |\mathcal{M}_1(\Pi_1|d_0, d_1)|^2 &= \left[\left| \mathcal{M}_R^{(1)}(\Pi_1) \right|^2 + 2 \Re \left(\mathcal{M}_R^{(1)}(\Pi_1) \mathcal{M}_{RV}^{(2)}(\Pi_1)^* \right) \right] \Theta(d(\Pi_1, \Pi_0^{\min}) - d_0) \\
 &\quad + \int \frac{d\Pi_2}{d\Pi_1} \left| \mathcal{M}_{RR}^{(2)}(\Pi_2) \right|^2 \Theta(d(\Pi_2, \Pi_0^{\min}) - d_0) \Theta(d_1 - d(\Pi_2, \Pi_1)) , \\
 |\mathcal{M}_2(\Pi_2|d_1)|^2 &= \left| \mathcal{M}_{RR}^{(2)}(\Pi_2) \right|^2 \Theta(d(\Pi_2, \Pi_1^{\min}) - d_1) .
 \end{aligned}$$

[Aside] Why the (s)EMD?

Any IRC-safe phase-space slicing d could have worked! mass, k_T , ...

But only a genuine metric has the **Triangle** and **Lipschitz Inequalities**.



Triangle

For any 3 events, we have:

$$d(\mathcal{E}, \mathcal{E}'') \leq d(\mathcal{E}, \mathcal{E}') + d(\mathcal{E}', \mathcal{E}'')$$

i.e. spheres are convex or “as small as possible”

Lipschitz

For (m)any IRC-safe observables, we have:

$$|\mathcal{O}(\mathcal{E}) - \mathcal{O}(\mathcal{E}')| \leq K d(\mathcal{E}, \mathcal{E}')$$

i.e. a metric is the *most different* two events can be! In an IRC-safe topology-preserving way

Bonus feature:

(s)EMD has the exact same form for *any* N-to-M particle configuration, particularly useful for NNLO

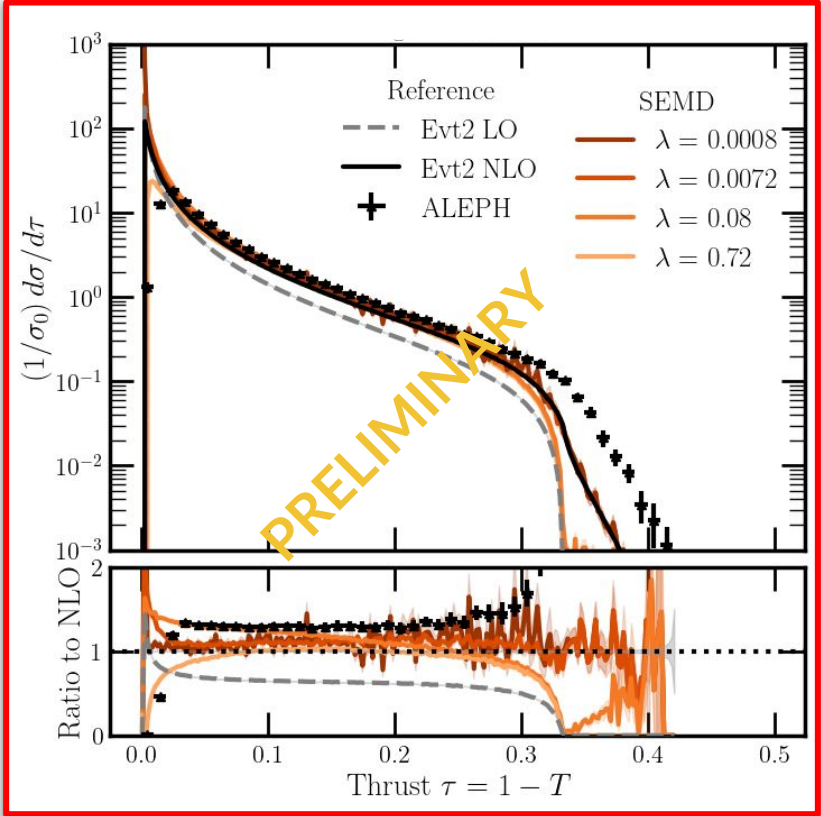
The (s)EMD is the “least violent” way to slice up phase space!

Works In Progress

- While the top two logs are universal, the non-universal parts in d are tricky. This is due to SEMD in the soft limit being tricky. We are still trying to calculate these! *This is the reason for all the “Preliminary” Tags throughout!*
 - This especially matters for NNLO, where there are now subleading logs that are non-universal.
- While the code is fast, it can be much faster. We hope to release this as a fully parallelized GPU-compatible JAX package.

Conclusion

- Using a metric to *smear* matrix elements is an effective way to mitigate negative weights and variance.
- Completely based in QCD, no ML or fits needed at all!
- Fast and efficient.
- Currently working on implementing NNLO, hopefully on arXiv soon!

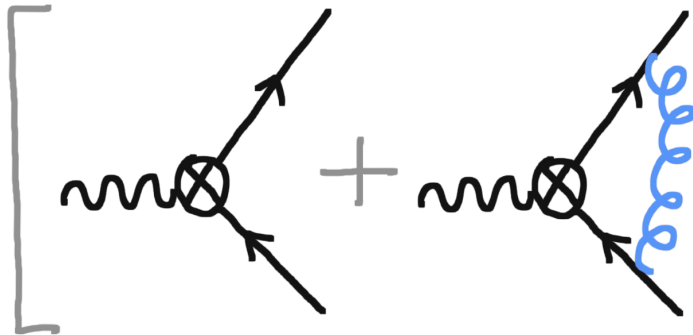


Appendices

Smearing

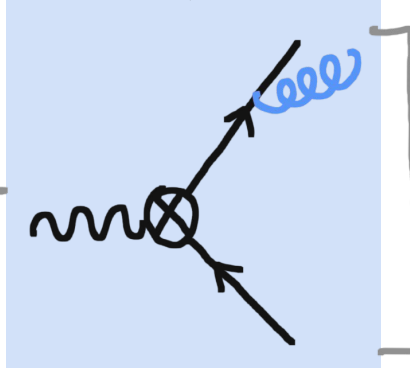
If a *resolved* event is close enough (within a distance d) of an unresolved event, treat it as *unresolved*.

Unresolved: Born + Virtual



Negative, Divergent

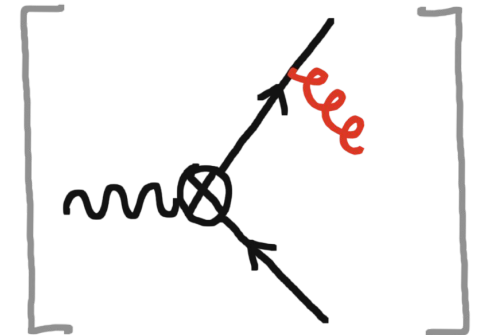
Real emission, close to Born



Positive, Divergent

Result: Finite & Indeterminate Sign

Resolved: Real Emission

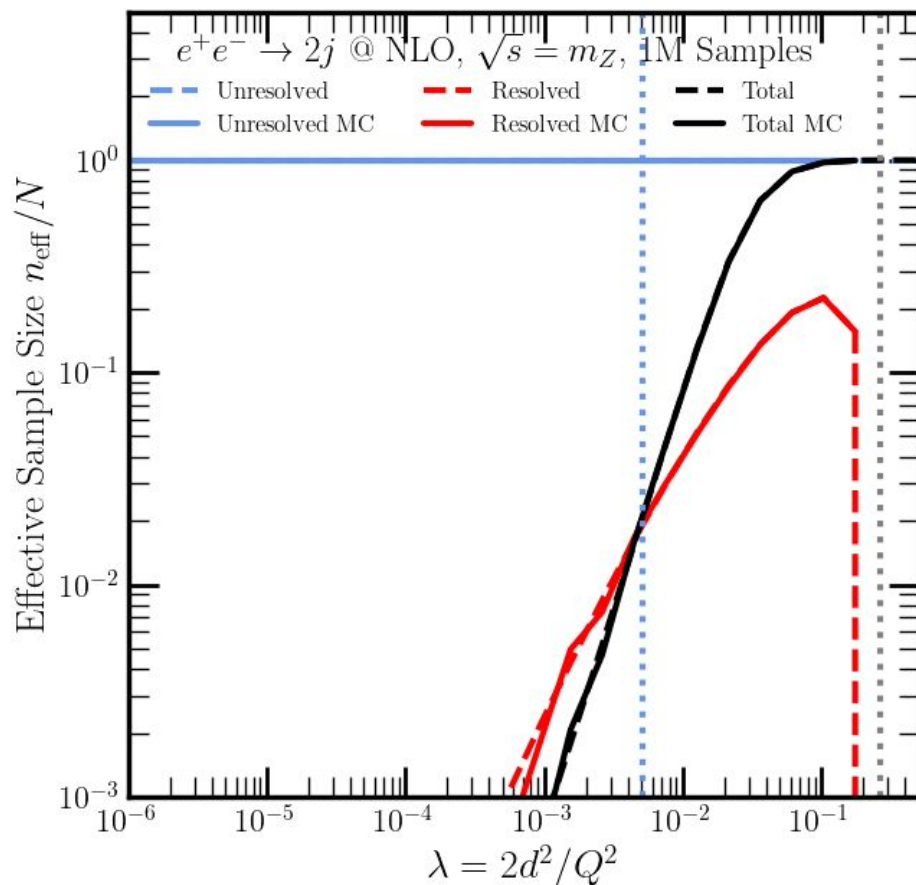


Positive, Finite

Negative for small d , positive for large d , and zero for d just right!

Phase Space Sampler Dependence

Uniform Phase Space (RAMBO)



Factorized Phase Space

